

Project Two Conference Presentation: Cloud Development
Video Link: <https://youtu.be/kHdJ6Ss7Vw>

Slide 2: Hello, my name is Shawn Millin. I am currently on the finishing end of my Computer Science degree at Southern New Hampshire University. Today I will be presenting the process of cloud development from the perspective of the Southern New Hampshire assignment in their Full-Stack II class, migration of a website to the cloud and AWS services. This will be presented so both technical and non-technical audiences can understand the process.

Slide 3: To migrate the application to the cloud the platform Docker was used to create containers. These containers each contain different parts of the application and their dependencies including the front-end and back-end of the application. The application for this project was a MEAN stack application where the data base was MongoDB, and used ExpressJS, Angular, and NodeJS.

Slide 4: The use of Docker Compose allowed the project's containers to communicate and interact with each other with a single command. This was possible by adding a YAML file to the application front-end and a YAML back-end. The Docker Compose allowed managing the application easier and updating easier than on a physical machine.

Slide 5: After experiencing Docker, this project moved on to migrating the application to Amazon Web Services (AWS). This is a serverless cloud computing model maintained by Amazon who also provides multiple serves for developers to use the cloud for applications. AWS provides a cloud service with a pay-as-needed model where a developer will only pay for the space their data takes up with virtually unlimited space. AWS takes care of the primary data security and maintenance so developers can focus on making applications without having a specialist handle maintenance.

The S3 service provided by AWS is where an application is uploaded, called a "bucket", and stored. These buckets are able to have more security added to them by giving special permissions to users in certain roles. A benefit of using this service is that the application can be accessed and updated anywhere there is an internet connection making collaborations seamless and all collaborators will be using the same platform instead of local machines.

Slide 6: The serverless cloud will scale automatically for a developer, both up and down. The AWS service of Lambda allows a developer to create functions for each action in an application. In this project's lambda functions were used to query a database through the Gateway API service. These functions were written in Python but other languages such as c#, node.js, Java, and in some places the JSON was used to write code.

Slide 7: For this project in AWS DynamoDB service was used in contrast to the MongoDB database used earlier for Docker. Both databases are document-based but DynamoDB used key-value pairs mainly with nested JSON-like documents and MongoDB is hierarchical in data structure within a single document.

Slide 8: A diagram of the traditional data centers' inability to automatically scale, causing firstly too much unused space then having a lack of space needed for scaling up.

Slide 9: AWS provides services to control access to the data of an application through their Identity and Access Management service (IAM). This allows developers to create different roles that users are allowed to utilize for authorization to certain application features. These are managed by codeing policies into the role or the application feature itself with JSON code. For this project a policy of making the web page public was done for the S3 bucket so public users can use the webpage's features

Secure connections were made between the services for this project's application features. The S3 bucket is secured by AWS automatically with the feature to allow public access to it. The connection security between the Lambda functions and the GatewayAPIs is the use of keys. The connection security between the Lambda functions and the DynamoDB database is through roles and policies.

Slide 10: The conclusion here based on this project is that cloud computing is a handy serverless service if developers choose to pay for storage in a pay-as-needed model, are wanting to focus on application making instead of devoting some time to server maintenance and security, need to create an application in a lower costing fashion, and need to scale up or down automatically.